Final Exam

Graded

Student

Zihan Zhou

Total Points

43 / 45 pts

Question 1

Overall Grades **43** / 45 pts

✔ **– 0 pts** See the details on each problem.

– 2 pts Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Print)	NetID	Signature
Zihan Zhou	zz451	周子涵

INSTRUCTION

- This is a 180-minute test.
- The total number of points is **45**.
- **No Cellphones.**
- **No Calculators.**
- **Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.**
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 周子涵

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	5	5	3	5	5	43

Problem 1. [5] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \qquad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$$

$\hat{R}_1 = \hat{R}_1 - 2\hat{R}_3$

$$\left[\begin{array}{ccc|c} 0 & -3 & -4 & -1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$$\Downarrow$$

$$\left[\begin{array}{ccc|c} 0 & 3 & 4 & 1 \\ 0 & 2 & 3 & -1 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$\hat{R}_1 = 2\hat{R}_1$

$\hat{R}_2 = 3\hat{R}_2$

$$\left[\begin{array}{ccc|c} 0 & 6 & 8 & 2 \\ 0 & 6 & 9 & -3 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$\hat{R}_2 = \hat{R}_2 - \hat{R}_1$

$$\left[\begin{array}{ccc|c} 0 & 6 & 8 & 2 \\ 0 & 0 & 1 & -5 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$\hat{R}_1 = \frac{1}{6}(\hat{R}_1 - 8\hat{R}_2)$

$$\left[\begin{array}{ccc|c} 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$\hat{R}_3 = \hat{R}_3 - 2\hat{R}_2$

$$\left[\begin{array}{ccc|c} 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \\ 1 & 0 & 0 & 11 \end{array}\right]$$

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

1)

$A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

$\Downarrow$

$\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

$\hat{R}_1 = \hat{R}_1 + \hat{R}_2$

$\hat{R}_4 = \hat{R}_4 + \hat{R}_2$

$\Downarrow$

$\begin{bmatrix} 0 & 2 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 3 \end{bmatrix}$

$\hat{R}_1 = \hat{R}_1 - 2\hat{R}_3$

$\hat{R}_4 = \hat{R}_4 - 3\hat{R}_3$

$\Downarrow$

$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$\Downarrow$

$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$Col(A) = \left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$.

2)

$Row_1 + 2Row_2$.

$Row(A) = \left\{ \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix} \right\}$.

(3)

$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$x - z = 0$

$y + z = 0$

let set $z = t$

then $x = t \quad y = -t$.

$v = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

basis of null space of A is $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = A \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

set $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix}$.

from the question we
ow that.

$b_1 - c_1 = 1 \quad a_1 - c_1 = 0 \quad -c_1 = 2$

$\Downarrow$

$c_1 = -2 \quad a_1 = -2 \quad b_1 = -1$

$b_2 - c_2 = 1 \quad a_2 - c_2 = 2 \quad -c_2 = 2$

$\Downarrow$

$c_2 = -2 \quad a_2 = 0 \quad b_2 = -1$

So

$$A = \begin{bmatrix} -2 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$$

Problem 4. [5] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

Set $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$.

find the eigenvalues.

$\det(A - \lambda I) = 0$

$(\lambda - 2)(\lambda - 1)^2 = 0$

$\lambda_1 = 1 \quad \lambda_2 = 2$.

$AM(1) = 2 \quad AM(2) = 1$.

for $\lambda = 1$

$A - \lambda_1 I = A - I$

$= \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix}$

$y = 0$

$-2x - 2y = 0$

$x = 0 \qquad z = 1$

$V_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

for $\lambda = 2$.

$A - 2I = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix}$

$-x + 2y = 0$

$-2x - 2y - z = 0$

set $z = 3t$.

$2y = -2x - 3t$.

$-x - 2x - 3t = 0$.

$x = -t$

$y = -\frac{1}{2}t$

$z = 3t$

$V_3 = \begin{bmatrix} -2 \\ -1 \\ 6 \end{bmatrix}$

find V_2 which satisfy

$(A - I)V_2 = V_1$

$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$2y = 0 \qquad y = 0$

$-2x - 2y = 1$

$-2x = 1$

$x = -\frac{1}{2}$

$z = t = 0$

$V_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$

Normalize vectors.

$V_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$V_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$

$V_3 = \begin{bmatrix} -2 \\ -1 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{-2}{\sqrt{41}} \\ \frac{-1}{\sqrt{41}} \\ \frac{6}{\sqrt{41}} \end{bmatrix}$

$P = \begin{bmatrix} 0 & -1 & \frac{-2}{\sqrt{41}} \\ 0 & 0 & \frac{-1}{\sqrt{41}} \\ 1 & 0 & \frac{6}{\sqrt{41}} \end{bmatrix} \qquad J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

$A = P \cdot J \cdot P^{-1}$

$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

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roblem 5. [5] Diagonalize the quadratic form

$$f(x, y, z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$f(x,y,z) = 3x^2+3y^2+4z^2+2xz-2yz$

$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$

$\lambda I - A = \begin{bmatrix} \lambda-3 & 0 & -1 \\ 0 & \lambda-3 & 1 \\ -1 & 1 & \lambda-4 \end{bmatrix}$

$\det(\lambda I - A) = 0$

$\det(\lambda I - A) = (\lambda-3)[(\lambda-3)(\lambda-4)-1] - (\lambda-3)$

$= (\lambda-3)[(\lambda-3)(\lambda-4)-2]$

$= (\lambda-3)[\lambda^2-7\lambda+10]$

$= (\lambda-3)(\lambda-2)(\lambda-5).$

$\lambda = 2 \quad \lambda_2 = 3 \quad \lambda_3 = 5.$

$\lambda = 2$

$A-2I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

$x+z=0$

find $(A-2I)V = 0$

$x+z=0$

$y-z=0$

$x-y+2z=0.$

let $z = t$

$x=-t \quad y=t$

$V_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$\lambda = 3$

$A-3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

$x-y+z=0$

$z=0$

let $y = t.$

$x = t.$

$V_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

③ $\lambda = 5$

$A-5I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix}$

$-2x+z=0$

$-2y-z=0$

$x-y-z=0$

let $z = -2t$

$y = t \quad x = -t.$

$V_3 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$

$P = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$

$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$A = PDP^{-1}$

$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

Problem 6. [5'] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

use an Augmented matrix

$$A = [v_1\ v_2\ v_3 \mid w]$$

$$= \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$\hat{R}_3 = \hat{R}_3 + R_2$

$$= \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$\hat{R}_2 \rightleftarrows \hat{R}_1$

$$= \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$\hat{R}_2 = \hat{R}_2 - 2\hat{R}_3$

$$= \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 0 & -3 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$\hat{R}_2 = -\frac{1}{3}\hat{R}_2$

$$= \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$\hat{R}_3 = \hat{R}_3 - \hat{R}_2$

$$= \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 \end{array}\right]$$

$\hat{R}_1 = (\hat{R}_1 + \hat{R}_3 - \hat{R}_2) \cdot \frac{1}{2}$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 \end{array}\right]$$

$x_1 = 3 \quad x_2 = -1$

$x_3 = 1$

$a = 3 \quad b = -1 \quad c = 1$

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Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

let $u_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ $u_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$

set v_1 which $u_1 \cdot v_1 = 0$.

$$v_1 = u_2 - \frac{u_1 \cdot u_2}{u_1 \cdot u_1} \cdot u_1$$

$$v_1 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2} \cdot \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$$

$$= \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}.$$ ✓

now $v_1 \cdot u_1 = 0$.

$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}\right\}$ −2

5/5

Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

we know that

$\det(A^T) = \det(A)$

and

$\det(kA) = k^n \det(A)$

since it times k to every row and $I_n = k\, I_n \Rightarrow \det(A) = k \det(A)$.

So $\det(kA) = k^n \det(A)$.

So.

$\det(A^T) = \det(-A)$

$\det(A) = (-1)^n \det(A)$

when n is odd.

$(-1)^n = -1$

which means.

$\det(A) + \det(A) = 0$

$\det(A) = 0$.

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Problem 9. [5] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

we can do the things below since symbol $\| \|$ means the distance which is always a Non-negative number.

$$\|\vec{u}+\vec{v}\|^2 = \vec{u}^2+\vec{v}^2+2\,\vec{u}\cdot\vec{v} = \vec{u}^2+\vec{v}^2+2\cdot\|\vec{u}\|\cdot\|\vec{v}\|\cdot\cos\theta \quad ①$$

$$\|\vec{u}-\vec{v}\|^2 = \vec{u}^2+\vec{v}^2-2\cdot\|\vec{u}\|\cdot\|\vec{v}\|\cdot\cos\theta \quad ②$$

When $\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$

$\Downarrow$

① = ②

minus $\vec{u}^2$, $\vec{v}^2$ from both side

we get $2\cdot\|\vec{u}\|\cdot\|\vec{v}\|\cdot\cos\theta = -2\|\vec{u}\|\cdot\|\vec{v}\|\cdot\cos\theta$.

$\vec{u}$, $\vec{v}$, $\vec{u}$ and $\vec{v}$

① $\vec{v}$ is zero vector.

then zero vector is perpendicular to every the vector. since $0\cdot\vec{u} = 0$ ($\vec{u}$ = any vector)

② $\vec{u}$ and $\vec{v}$ is not zero vector.

then we get

$4\|\vec{u}\|\cdot\|\vec{v}\|\cdot\cos\theta = 0$.

and $\|\vec{u}\|\cdot\|\vec{v}\|\cdot 4 \neq 0$.

So $\cos\theta = 0$.

Since $\theta \in [0, 2\pi]$

$\theta = \frac{\pi}{2} / \frac{3\pi}{2}$.

two answer will show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.